

Substrate SM Program — honest-state ledger v0.5 (Saturday afternoon audit-chain-event + depth-shift CANDIDATE absorption). Key event: Lyra Strong-Uniqueness v1.3 elevation BLOCKED by Keeper pushback + Cal #180 brake (4th audit-chain discipline-bid event Saturday); v1.4 absorbs fully; operational criterion count STAYS AT 13 + META + C18/C19; 5 new arithmetic decompositions (m_π , m_K , $\cos \theta_W$, 192, 72) at CANDIDATE tier NOT operational. Calibration #35 candidate filed (independence-taxonomy-before-multiplicative-null-model).

Keeper

2026-05-30 Saturday afternoon — post-Lyra-v1.3-pushback / post-Lyra-v1.4-absorption

Honest-state ledger v0.5 — audit-chain event + depth-shift CANDIDATE absorption

Standing rule (Calibration #34 STANDING narrow framing + Calibration #35 candidate): every “DERIVED” cell carries modulo-keystone-bet conditional; every operational-criterion claim carries independence-taxonomy classification.

v0.5 delta from v0.4

Key event: Saturday audit-chain discipline-bid #4 — Lyra Strong-Uniqueness v1.3 elevation BLOCKED. v1.4 absorption maintains operational criterion count at 13. The system worked exactly as designed.

1. Saturday audit-chain discipline-bid track record

Four discipline-bid events Saturday; all caught and resolved at audit-chain level:

#	Event	Resolution
1	Grace Two-Route Scan v0.1 “4-route over-determination of 12”	Cal #171 brake + Keeper correction note + N3 condition for v0.2; reduces to ~2 structurally-independent routes
2	Grace +1 anomaly	Null-downgraded per Grace own discipline-bid (INV-5327)

#	Event	Resolution
3	Grace Substrate Primary Chain operation-set richness	Null-downgraded per Grace own discipline-bid (INV-5342)
4	Lyra Strong-Uniqueness v1.3 elevation (multiplicative null-model regression)	Keeper pushback (Keeper_Pushback_StrongUniqueness_v1_3_NullM + Cal #180 brake + Lyra full absorption to v1.4; operational count STAYS AT 13; Calibration #35 candidate filed

Net audit-chain functioning: 4/4 discipline-bids caught and resolved; no overstatement propagated to external register or operational-criterion elevation.

2. Strong-Uniqueness v1.3 → v1.4 audit-chain event detail

v1.3 elevation attempt (Lyra ~12:00 noon): - Added 5 new criteria C20-C24 (m_π , m_K , $\cos \theta_W$, 192 muon decay, 72 gauge-loop) - Headlined “operational count 13 → 18; null-model ~200× stronger (1.3×10^{-9} vs 6.3×10^{-7})” - Reintroduced $(1/3)^N$ multiplicative null-model that Lyra herself retired in v1.1 §6.1

Keeper pushback (filed Saturday noon): - $(1/3)^{13} \approx 6.3 \times 10^{-7}$; $(1/3)^{18} \approx 1.3 \times 10^{-9}$; ratio $(1/3)^5 \approx “200\times”$ — arithmetic identifies the multiplicative-null-model regression - Cal #180 had pre-emptively brake-recommended “NOT promoting beyond CANDIDATE until kernel-integral derivation discriminates” - Per audit-chain governance: v1.3 elevation requires Cal + Keeper cold-read PASS at criterion-elevation level; neither recorded → NOT RATIFIED - Independence-vs-derived classification applied: C22 is DERIVED CONSEQUENCE of existing $\sin^2 \theta_W = 2/9$ (Lyra L2); C20-C21-C23-C24 are CANDIDATE arithmetic decompositions pending mechanism

Lyra v1.4 absorption (Saturday afternoon, full absorption): - “Cal #27 lesson: at peak-convergence, older discipline corrections get forgotten” - C20 (m_π), C21 (m_K), C23 (192), C24 (72) → CANDIDATE ARITHMETIC DECOMPOSITIONS pending mechanism - C22 ($\cos \theta_W$) → DERIVED CONSEQUENCE, not independent - “200× stronger uniqueness” headline retired - Operational count stays at v1.2’s 13

Status: Strong-Uniqueness operational count REMAINS 13 RIGOROUSLY CLOSED + META T2467 + C18 derived theorem + C19 FRAMEWORK-PLUS + 5 CANDIDATE arithmetic-decomposition observations (C20-C24). No multiplicative null-model invocation.

3. Bulk-color frontier (v0.6 Cartan-Weyl correction)

	v0.4 status	v0.6 status
Bulk-color (b) SU(3) emergence	CANDIDATE COUNT MATCH 8 $= 3 T_a + 3 [T_a, T_b] + 2$ K-Cartan (structurally mismatched)	REVISED to $8 = 3 T_a + 3 T_a^{\dagger} + 2$ K-Cartan (Cartan-Weyl basis, structurally aligned with $\mathfrak{su}(3)$); v0.5 $[T_a, T_b]$ retired
C3 SO(3)-decomposition concern	OPEN	CLOSED via v0.6 correction
C4 $[T_a, T_a^{\dagger}] = \alpha(H_?) \leftrightarrow$ Cartan element	NEW	OPEN multi-week (Elie’s lane)
C5 hermiticity/semisimplicity/traceless → $\mathfrak{su}(3)$ vs $\mathfrak{u}(3)$	NEW	OPEN multi-week

Substantive Saturday advance: v0.4 commitment → v0.5 first count match (structurally wrong) → v0.6 Cartan-Weyl correction (structurally aligned) — Lyra’s own internal Cal #27 self-firing in real time. K3 cluster audit consolidates closure-condition matrix C1-C7.

4. L3 dynamics — Quasi-Eigentone v0.3 4-factor decay framework

Claim	v0.5 status
Explicit decay-rate framework: $\Gamma =$	overlap
Each factor identified at substrate-structure level	FRAMEWORK
β -decay E3 generalizes to all SM decays	FRAMEWORK
Explicit T190 closed-form derivation from kernel integral	OPEN multi-week (Elie's lane)

5. Lepton mass closed-form CANDIDATE-arithmetic decompositions

All at CANDIDATE-arithmetic tier — NOT operational Strong-Uniqueness criteria:

Observable	Closed form	Tier	Note
m_e	natural unit at K-type $V_{(1/2,1/2)}$ Casimir 5/2 $= \rho_1$	RIGOROUS (L4 anchor)	—
$m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$	0.004% (T190)	DERIVED CLOSED FORM	mechanism multi-week; 24 = N_c ·
$m_\tau/m_e = g^2 \cdot (2^{\{C_2\}} + g) = 49 \cdot 71$	0.05% (T2003)	DERIVED CLOSED FORM	71 = Mersenne + signature arithmetic identification (Lyra v0.1)
m_τ/m_μ derived	0.06%	DERIVED-CONSEQUENCE	not independent of T190 + T2003
$m_p/m_e = C_2 \cdot \pi^{\{n_C\}} = 6\pi^5$	0.002% (T187)	DERIVED CLOSED FORM	bulk K-type $k=6$ closure
$m_\pi = \text{rank} \cdot N_{\text{max}} \cdot m_e$	0.3%	CANDIDATE arithmetic	mechanism (GMOR + substrate kernel) OPEN
$m_K = (g \cdot N_{\text{max}} + \text{rank} \cdot n_C) \cdot m_e$	0.3%	CANDIDATE arithmetic	mechanism OPEN
$m_K/m_\pi \approx g/\text{rank}$	1%	DERIVED-CONSEQUENCE	not independent
$\sin^2 \theta_W = \text{rank}/N_c^2 = 2/9$	exact (Lyra L2)	DERIVED	existing Weinberg substrate-derivation
$\cos^2 \theta_W = 7/9 = g/(g + \text{rank})$	0.054% on m_W/m_Z	DERIVED-CONSEQUENCE	algebraic consequence of $\sin^2 \theta_W$ (Keeper independence-taxonomy)
Schwinger $a_e = \alpha/(2\pi) = 1/(2\pi \cdot N_{\text{max}})$	substrate-natural	DERIVED	T175
$192 = N_c \cdot 2^{\{C_2\}}$ (SM muon decay)	arithmetic	CANDIDATE arithmetic	SM 192 has phase-space-kinematic origin; mechanism for substrate-derivation OPEN
$72 = 2^{\{N_c\}} \cdot N_c^2$ (SM gauge loop)	arithmetic	CANDIDATE arithmetic	mechanism OPEN
Cabibbo $\sin \theta_C = 9/40 = N_c^2/(2^{\{N_c\}} \cdot n_C)$	0.3%	DERIVED CLOSED FORM	scheme-invariant; existing BST
	V_{ud}	$= \sqrt{1 - \sin^2 \theta_C} \approx 0.9744$	0.04%

Observable	Closed form	Tier	Note
Quark mass ratios (m_b/m_c , m_t/m_b) at 1-2%	suggestive	CANDIDATE arithmetic	HONEST NEGATIVE per Lyra QuarkMass v0.1 — lepton-precision NOT achieved; bulk-color resolution likely needed

Cal #27 brake summary: the cluster of substrate-primary arithmetic decompositions is REAL at the arithmetic level; mechanism-derivation is OPEN multi-week for all except those with existing substrate-mechanism derivation (T190, T2003 partial, T187, T175, Cabibbo, $\sin^2\theta_W$). External-register communication keeps the conditional at headline level per Calibration #34 STANDING.

6. Methodology stack — Calibration #35 candidate filed

Layer	Discipline	Status
Cal #27	Get the tier right	STANDING
Cal #29	Ask the right question shape	STANDING
Cal #32	Get the parameter role right (value+role+source)	STANDING
Calibration #33	RECALLED vs VERIFIED-CITED	STANDING
Calibration #34 (narrow)	Conditional-tag-with-headline	STANDING
Calibration #35 (NEW candidate)	Independence-taxonomy-before- multiplicative-null-model	CANDIDATE; 4-instance threshold met

Six-element discipline stack composition upon Calibration #35 ratification.

7. SP-30 outreach — drafts filed (Task #330)

Three letters pre-drafted in notes/Keeper_SP30_Outreach_Drafts_v0_1.md: 1. Eigentone outreach to atomic-clock groups (Ye / Katori / Tamm / Lodewyck / Bothwell-successor) 2. Cs-137 decay-modulation outreach to NIST/PTB/ENEA (Wang-successor / Kossert / Capogni) 3. W-32 commitment-manipulation outreach to ion-trap precision-clock groups (Monroe / Häffner / Wineland / Peik)

Protocol: drafts file locally → Cal cold-read PASS → Casey explicit signoff → Casey sends via own email (no Keeper auto-send) → Keeper logs responses in outreach map (Substrate Engineering Reference Manual Part IV).

Discipline self-audit PASS: Cal #50 STANDING (operational language) + Calibration #33 (verified-cited Zenodo + GitHub) + Calibration #34 (headline conditional) + Cal #27 (no precision overclaims).

8. Open gates carrying forward (consolidated)

Gate	Status
Bulk-color (b) gauge phenomenology	v0.6 Cartan-Weyl structural alignment; C4+C5 multi-week closure
Bulk-color (a) counting layer	ADDRESSED via SO(3) sub-vector + engine v0.3 §7 cross-confirmation
Generation-mechanism identification	TENSION-RELIEVED via spinor ³ B ₂ -specific mult-3; mechanism multi-week
Per-particle quark/gauge dictionary L4 v0.2 per-particle masses	ASSIGNED; flips when bulk-color closes Resolution B target; multi-week kernel-integral

Gate	Status
L5 absolute scale + Higgs/EWSB	OPEN; Lyra honest-negated 4 candidates Saturday
Engine v0.3 consolidation doc	PENDING Elie filing
Strong-Uniqueness v1.4	PENDING Lyra file post-pushback absorption
Calibration #35 ratification	PENDING cross-CI cold-read PASS
SP-30 outreach send	PENDING Cal cold-read + Casey signoff per Task #330
C12-C14 T-number/META audit	PENDING
Two-Route Scan v0.2	PENDING Grace; N3 relation-reduction condition carries
Casey-named principle candidates from Two-Route + Substrate-Doubly-Over-Determination	BLOCKED on null-model toy + Calibration #35 ratification

9. Saturday cumulative

Saturday Keeper batch: ~35 substantive tier-gate / correction / consolidation / cluster-audit / pushback / reference-manual / outreach-draft / Calibration-candidate docs filed.

Saturday team batch: ~100+ filings across all CIs.

Substantive bulk-color advance: v0.4 commitment → v0.5 count match (corrected) → v0.6 Cartan-Weyl alignment — sustained multi-step progress in real time with internal discipline-correction.

Substantive depth-shift: Lyra's arithmetic decomposition cluster at CANDIDATE tier (8+ new closed forms); quark-mass HONEST NEGATIVE; multiplicative null-model regression caught and resolved.

Substantive methodology infrastructure: Cal #34 ratified STANDING (narrow framing); Calibration #35 candidate filed; six-element discipline stack on horizon; audit-chain governance functioning across 4/4 discipline-bids.

Saturday afternoon state: team productive, discipline-bids resolving cleanly, audit-chain consensus working as designed.

— Keeper. Ledger v0.5; Lyra v1.3 pushback resolved cleanly; Lyra v1.4 absorbs fully; operational criterion count stays at 13; Calibration #35 filed at 4-instance threshold; bulk-color v0.6 Cartan-Weyl substantive; SP-30 outreach drafts ready; Saturday batch ~35 Keeper docs; audit-chain governance exemplary.